Computer Networks Lab 10

Course: Computer Networks (CL3001) | Semester: Fall 2026
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Learning Objectives

- Introduction of OSPF.
- Introduction of BGP.
- To verify OSPF routing and test connectivity between routers using ping commands.
- To understand the basics of BGP and configure it between different Autonomous Systems to verify route advertisement.

Open Shortest Path First (OSPF)

OSPF works by using the Dijkstra algorithm. First, a shortest path tree is constructed, and then the routing table is populated with the resulting best paths. OSPF converges quickly, although perhaps not as quickly as EIGRP, and it supports multiple, equal-cost routes to the same destination. Like EIGRP, it does support both IP and IPv6 routed protocols.

OSPF Terminology:

- Router ID.
- Neighbor.
- Adjacency.
- Hello protocol.
- Neighborship database.
- Link State Advertisement.
- Topological database.
- OSPF areas.
- Loopback address.
- Link costs.

OSPF provides the following features:

- Consists of areas and autonomous systems.
- Minimizes routing update traffic.
- Allows scalability.
- Supports VLSM/CIDR.
- Has unlimited hop count.
- Allows multi-vendor deployment (open standard).

1. OSPF Practical Part

A subnet is just a range of IP addresses. All the devices in the same subnet can communicate directly with one another without going through any routers. In IPv4, a network interface is connected to only one subnet and has only one IP address. In IPv6 things are slightly more complicated, so we'll save IPv6 subnetting for another article. But it's useful to understand IPv4 first because the basic concepts are the same.

Lab_10#config t Lab_A(config)#router ospf 1

1: is OSPF process number: out of scope for CCNA, range: <1-65535>

Lab_10(config-router)#network 10.0.0.0 0.255.255.255 area 0 0.255.255.255 : an example of wildcard

Wildcard

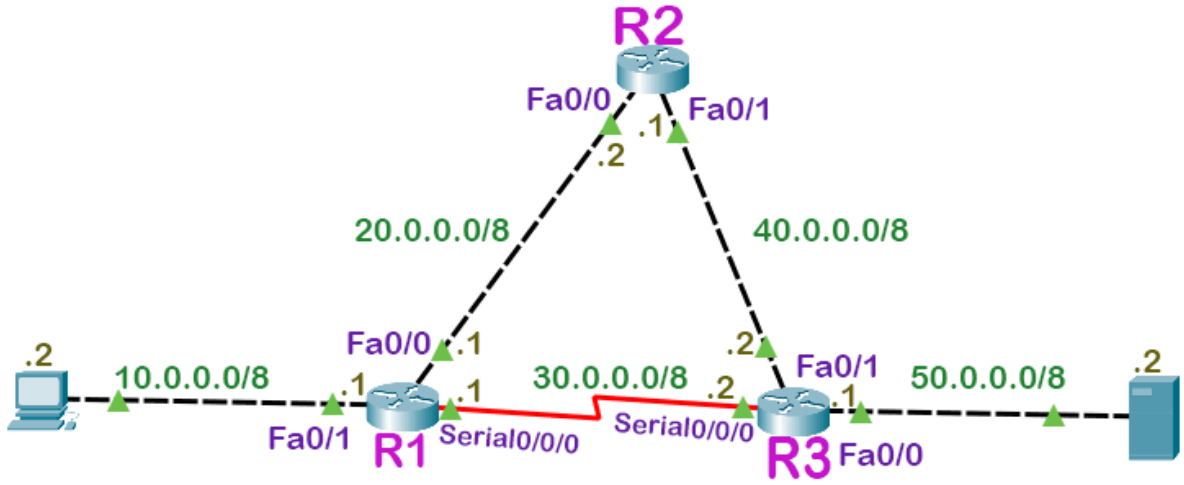
The wildcard mask can be configured as the inverse of a subnet mask. For example, IP 172.16.1.16/28 network. The subnet mask for this interface is /28 or 255.255.255.240. The inverse of the subnet mask results in the wildcard mask.

255.255.255.255
- 255.255.255.240 (Subtract the subnet mask)
---
0. 0. 0. 15 Wildcard mask

255.255.255.255
255.255.255.0 (Subtract the subnet mask)
---
0. 0. 0. 255 Wildcard mask

255.255.255.255
255.255.0.0 (Subtract the subnet mask)
---
0. 0. 255. 255 Wildcard mask

2. Implementation of OSPF



Test connectivity from R1

```
Router>en
Router#ping 10.0.0.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.0.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

Router#ping 20.0.0.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 20.0.0.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/0 ms

Router#ping 30.0.0.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 30.0.0.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 11/11/12 ms
```

Like this test connectivity from R2 and R3.

OSPF Routing commands

The following two steps in global configuration mode enable OSPF routing on the router.

1. Enable an OSPF process.
2. Assign an area to the interfaces.

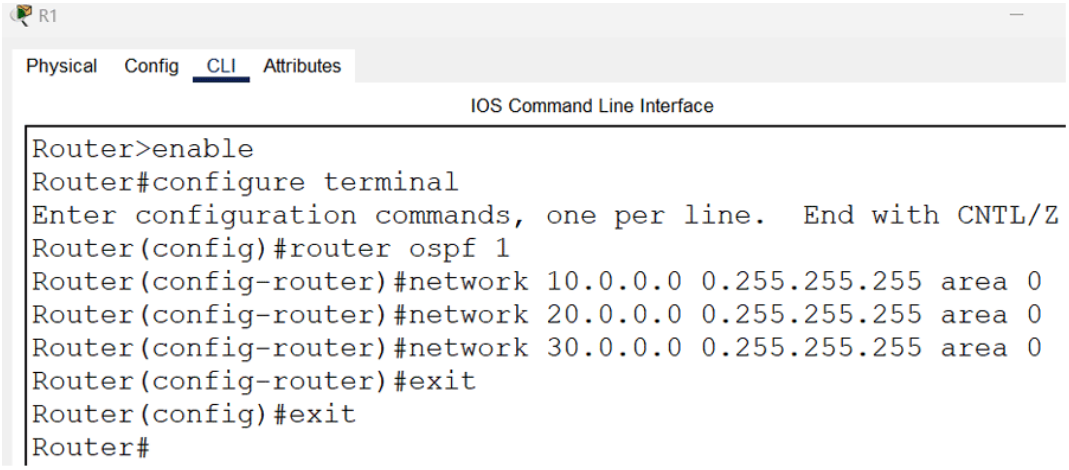
The following command enables an OSPF process on the router.

Router(config)# router ospf [process_ID]

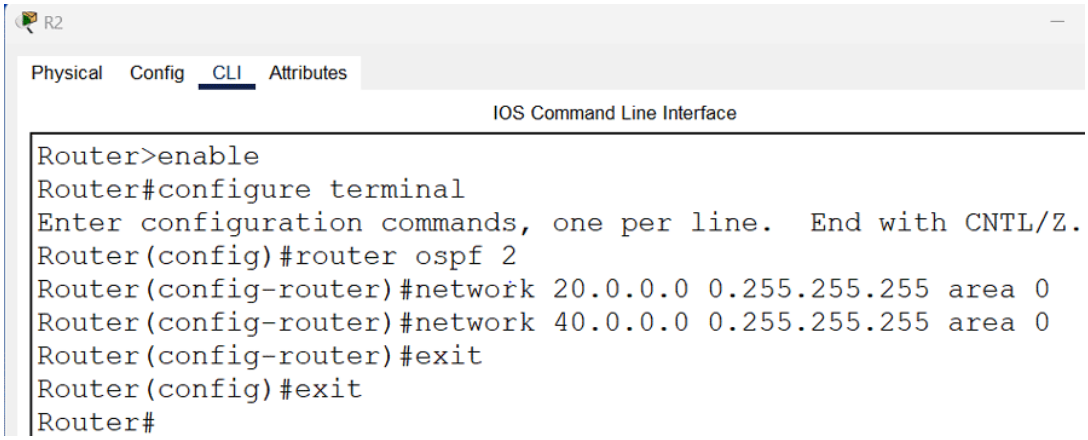
The following command assigns an area to the interface.

Router(config-router)# network [IP_network_#] [wildcard_mask] area [area number]

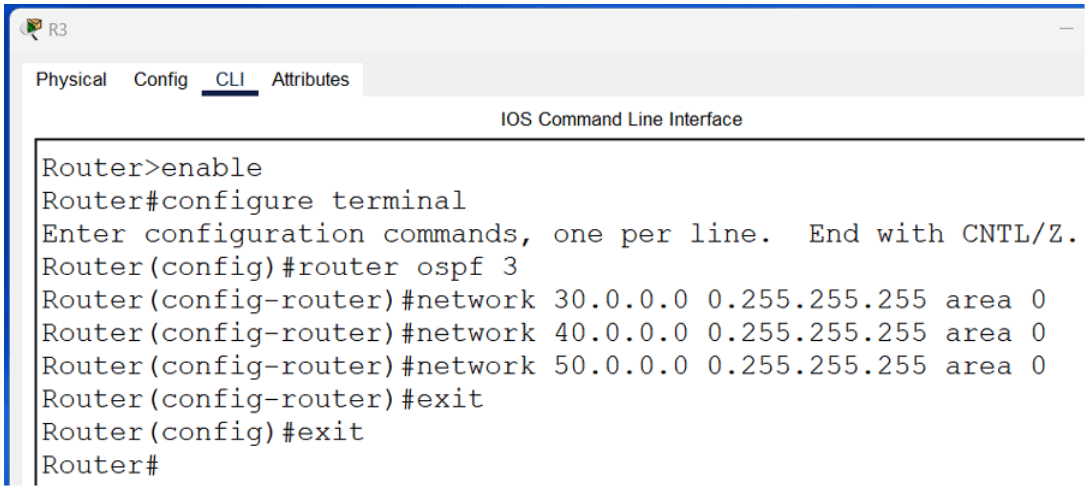
Enabling OSPF Routing on R1



Enabling OSPF Routing on R2



Enabling OSPF Routing on R3



Now, verify the OSPF routing in all routers:

```
Router#sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

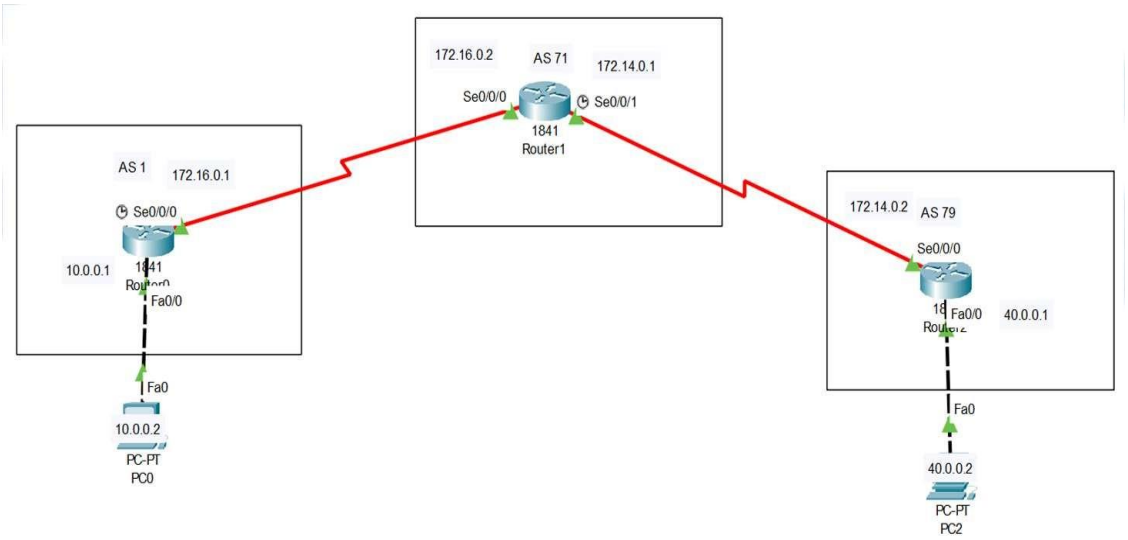
Gateway of last resort is not set

O    10.0.0.0/8 [110/3] via 40.0.0.1, 00:08:36, FastEthernet0/1
O    20.0.0.0/8 [110/2] via 40.0.0.1, 00:08:36, FastEthernet0/1
C    30.0.0.0/8 is directly connected, Serial0/0/0
C    40.0.0.0/8 is directly connected, FastEthernet0/1
C    50.0.0.0/8 is directly connected, FastEthernet0/0
```

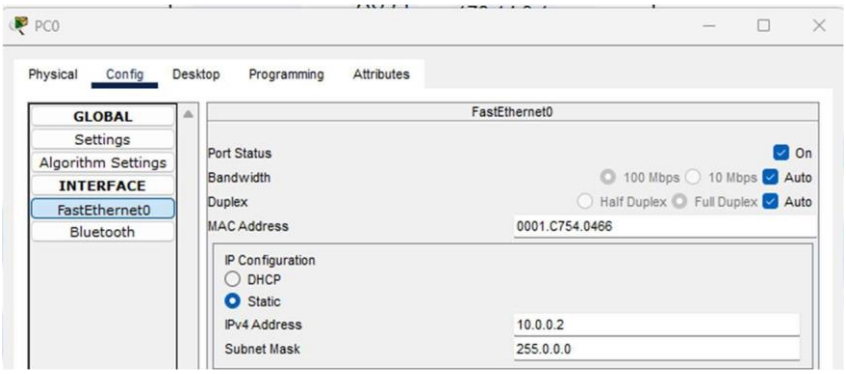
Border Gateway Protocol (BGP)

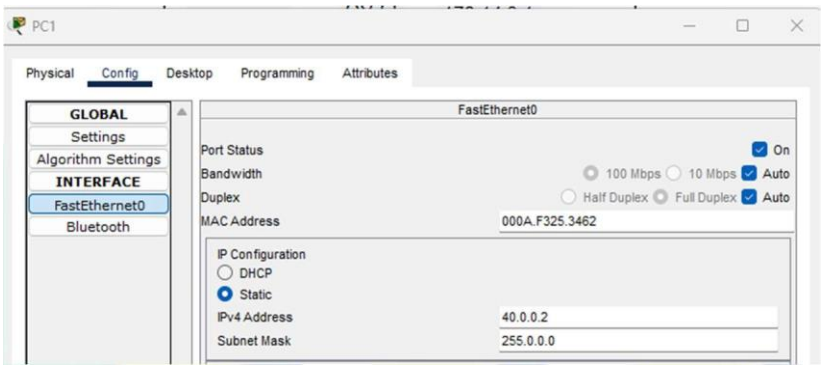
Border Gateway Protocol (BGP) is a true path-vector routing protocol. BGP maintains a table of network paths and makes routing decisions based on path, network policies, and rule sets. It is primarily used for routing between autonomous systems (AS) on the internet. Instead of relying on hop count, BGP considers multiple attributes like AS path, next-hop, and local preference to determine the best route. BGP ensures stable and scalable inter-domain routing but requires careful configuration and management to prevent routing loops and optimize performance.

Below figure shows a topology with BGP:



Assign IPs on PCs:





Configuration on Router 0:

R0(config)#router bgp 1

R0(config-router)#neighbor 172.16.0.2 remote-as 71

R0(config-router)#network 10.0.0.0 mask 255.0.0.0 R0(config-router)#exit

R0(config)#do write Building configuration... [OK]

Configuration on Router 1:

R1(config)#router bgp 71

R1(config-router)#neighbor 172.16.0.1 remote-as 1

R1(config-router)#%BGP-5-ADJCHANGE: neighbor 172.16.0.1 Up

R1(config-router)#neighbor 172.14.0.2 remote-as 79

R1(config-router)#network 40.0.0.0 mask 255.0.0.0

R1(config-router)#exit

R1(config)#do write

Building configuration...

[OK]

Configuration on Router 2:

R3(config)#router bgp 79

R3(config-router)#neighbor 172.14.0.1 remote-as 71

R3(config-router)#%BGP-5-ADJCHANGE: neighbor 172.14.0.1 Up

R3(config-router)#network 40.0.0.0 mask 255.0.0.0 R3(config-router)#exit

R3(config)#do write

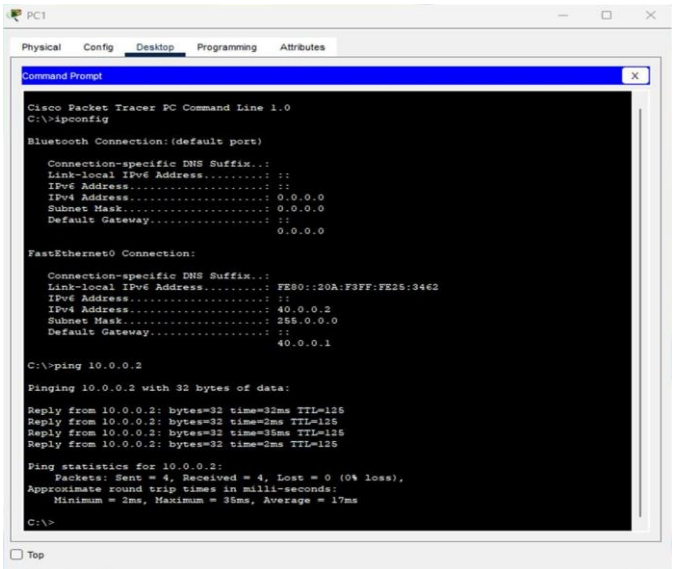
Building configuration...

[OK]

BGP Testing & Troubleshooting

For BGP testing we will ping both pc and check the network communication.

On PC1:



Check BGP route on router R1:

```
R1>enable
R1#
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#interface FastEthernet0/1
R1(config-if)#
R1(config-if)#exit
R1(config)#interface FastEthernet0/0
R1(config-if)#exit
R1(config)#
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

C    10.0.0.0/8 is directly connected, FastEthernet0/0
B    40.0.0.0/8 [20/0] via 172.16.0.2, 00:00:00
C    172.16.0.0/16 is directly connected, Serial0/0/0

R1#
```

Check whether BGP protocols configured or not on Router:

```
R1>en
R1#show ip protocols
Routing Protocol is "bgp 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  IGP synchronization is disabled
  Automatic route summarization is disabled
  Neighbor(s):
    Address          FiltIn FiltOut DistIn DistOut Weight RouteMap
    172.16.0.2
  Maximum path: 1
    Routing Information Sources:
      Gateway         Distance      Last Update
      172.16.0.2              20          00:00:00
    Distance: external 20 internal 200 local 200
R1#
```

Show BGP Status:

```
R1>en
R1#show ip bgp summary
BGP router identifier 172.16.0.1, local AS number 1
BGP table version is 5, main routing table version 6
2 network entries using 264 bytes of memory
2 path entries using 104 bytes of memory
1/1 BGP path/bestpath attribute entries using 184 bytes of memory
2 BGP AS-PATH entries using 48 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
Bitfield cache entries: current 1 (at peak 1) using 32 bytes of memory
BGP using 632 total bytes of memory
BGP activity 2/0 prefixes, 2/0 paths, scan interval 60 secs

Neighbor        V    AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
172.16.0.2      4    71     146     140        5    0    0 01:15:45        4

R1#
```

Show BGP neighbor's status:

```
R1>en
R1#show ip bgp neighbors
BGP neighbor is 172.16.0.2,  remote AS 71, external link
  BGP version 4, remote router ID 172.16.0.2
  BGP state = Established, up for 01:16:49
  Last read 01:16:49, last write 01:16:49, hold time is 180, keepalive interval is 60
seconds
  Neighbor capabilities:
    Route refresh: advertised and received(new)
    Address family IPv4 Unicast: advertised and received
  Message statistics:
    InQ depth is 0
    OutQ depth is 0

                         Sent       Rcvd
    Opens:                  2          2
    Notifications:          0          1
    Updates:                2          4
    Keepalives:           139        140
    Route Refresh:          0          2
    Total:                143        149
  Default minimum time between advertisements runs is 30 seconds

 For address family: IPv4 Unicast
  BGP table version 5, neighbor version 6/0
  Output queue size : 0
  Index 1, Offset 0, Mask 0x2
  1 update-group member
                               Sent       Rcvd
  Prefix activity:             ----       ----
    Prefixes Current:             1          1 (Consumes 46 bytes)
    Prefixes total:               1          1
    Implicit Withdraw:            0          0
    Explicit Withdraw:            0          0
    Used as bestpath:           n/a          1
    Used as multipath:          n/a          0
```

Similarly, we can check the BGP route on every other Routers as well by verifying the advertised and received routes. This helps ensure proper path selection and confirms that routing policies are applied correctly.